Nimish Wadhwa

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.91
Sant Gyaneshwar Model School , Alipur, Delhi	Class XII (CBSE)	2022	91.80%
Ryan International School , Mayur Vihar , Delhi	Class X (CBSE)	2024	96.20%

PROJECTS

AttendX - Face Recognition Attendance Platform

- Built a real-time face recognition attendance system using **YuNet** for fast face detection and **FaceNet512** embeddings for identity representation, with **FAISS** enabling **sub-second** similarity search across all enrolled users
- Eliminated cross-tenant data leakage by architecting isolated per-organization **FAISS indexes** with **JWT-based auth**
- Deployed end-to-end with **FastAPI**, **Docker**, and **Supabase**, hosted live on Hugging Face Spaces.
- Enabled thread-safe **concurrent attendance marking** via per-classroom **RW locks** on shared FAISS indexes.

Delhi Weather Forecasting -- End-to-End ML System

- Built a **multi-output CNN-LSTM** model to forecast temperature and rainfall 6 hours ahead, trained on **20 years** of hourly meteorological data spanning **175K+ records**, capturing long-term seasonal and short-term weather patterns
- Engineered **70+ features**, including cyclical time encodings and rolling statistics, and fine-tuned the model on rolling 720-hour windows with frozen convolutional layers to prevent catastrophic forgetting

Credit Risk Prediction Model

- Built an **XGBoost** model for loan default prediction on **148K+ records**, addressing data leakage from post-approval fields and narrowing **30+** features to **18** key predictors via Random Forest importance ranking.
- Benchmarked **SMOTE** vs. **gradient-scaling** for **class imbalance,** tuning via **Stratified K-Fold CV,** and optimized the decision threshold to achieve **86.0% accuracy and 0.84 ROC-AUC** score **.**
- Applied **SHAP** analysis for model interpretability, surfacing the top drivers behind individual default predictions.

SKILLS

- **Programming Languages** : Python , C++ , SQL
- **Libraries** : Numpy , Pandas , Matplotlib , Tensorflow , Keras , Scikit-Learn , Langchain , LangGraph
- **Domains** : Machine Learning , DeepLearning , Transformers ,Time Series Analysis , DSA , OOPS
- **Tools** : Git , Github , FastAPI , Docker , PowerBI , HuggingFace

- **Business :** Stakeholder Engagement, Cross-Functional Collaboration, Initiative Ownership, Structured Problem Solving

ACHIEVEMENTS

- **KLA AI Hackathon** - Ranked among the **Top 5 teams** out of **100+ participants** by developing a semiconductor image restoration model that achieved the highest competition accuracy **(89.75%)**.

- **Deutsche Börse Cybersecurity Hackathon** - Secured **2nd Place** by solving cybersecurity **CTF (Capture the Flag)** challenges and delivering a structured gap assessment report.

- **Amazon ML Summer School 2026** - Earned selection among the top **3,000** out of **134,000+ applicants**.

- Solved **400+ DSA problems** and achieved a Codeforces rating of **1420**, earning the **Specialist** title

RELEVANT COURSES

Programming and Machine Learning Courses - Introduction to Programming in C language, Introduction to AI/ML

Mathematics - Linear Algebra, Applied Maths, Calculus , Numerical Methods , Differential Equation

POSITIONS OF RESPONSIBILITY

Internship Coordinator | Office of Career Services

- Identified and engaged **30+ prospective companies** through a structured outreach strategy, tailoring each stakeholder pitch to organizational hiring needs, and successfully onboarded a new recruitment partner strengthening OCS's corporate network

Consulting Coordinator | Finance and Consulting Club , IIT Hyderabad

- Drove **5,000+** footfall for Horizon, IIT Hyderabad's flagship management fest, by leading a **10-member** associate team to architect high-engagement competitions and curated event experiences.

Class Representative | Btech Chemical Engineering

- Elected by a batch of **40 students** to lead as the official liaison between students and faculty, managing communication and representing peer interests in academic matters.

CERTIFICATIONS

- **Convolutional Neural Networks** : DeepLearning.AI
- **Neural Networks** : DeepLearning.AI
- **Advanced Learning Algorithms** : DeepLearning.AI

- **Data Analysis with Python** : FreeCodeCamp

- **McKinsey Forward Fellow**

EXTRACURRICULAR

- **Gold Medal** -- Football, Diesta (Inter-Department Sports Fest, IIT Hyderabad)
- **Bronze Medal** -- Football, Milan (Inter-Hostel Sports Fest, IIT Hyderabad)